"Instead Of Replacing Current Leaders In Terms Of SEMICONDUCTORS, We Should TRY AND FIND SOLUTIONS To Newer Problems"

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Krishnan Shrinivasan, managing director at Lam Research India, in an exclusive interaction with Mukul Yudhveer Singh spoke about whether India should invest in a semiconductor fab or not. Here is what he had to say

Does India require a semiconductor fab?

Not just India, I think every major economy should have access to a spectrum of technologies. The question should not be whether or not we should have a fab, but how soon can we have a semiconductor fab? Should it be set up by private players, or should the government play a role, or should it be a mix of both, are the right questions to ask!

The government's Chandigarh fab is a proof that capabilities exist in India. After the capability comes the question of economic benefit. Unless the answer to this question is clear, private players may not be interested in investing in a semiconductor fab. The good news is that a lot of multinational companies have started coming to India to get their electronic products assembled here.

I think it is just a matter of time before India moves up the value chain in terms of electronics manufacturing. The journey may be from designing circuit boards to their assembly, to manufacturing them, and so on, but it is just a matter of time. Once we have done that it will become attractive for a lot of big players to set shop in India.

By when should India have a semiconductor fab?

We missed that boat about forty years ago. A truth that not many know is, that amongst the Asian nations, India had the fab development going on earliest. Even countries like Japan were having the same level of discussions around fabs in the 1980s, just like us in India.

I wish we move fast now because everything in the ecosystem outside semiconductor fab already exists in India. The biggest proof of the same is that every big chip manufacturing company of the world has a technical office in India. Then there are a good number of startups designing chips in India that have the potential to go global. To top it all, we have a large base to consume it all here in the country. If we just take the example of smartphones, there are hundreds of millions of consumers waiting to upgrade their smartphones.

Then we have a huge consumption of consumer electronics. The defense, aerospace, and space sectors also use a lot of semiconductors. Most importantly, the level of human talent available in the country is excellent. The number of Indians working in the semiconductor industry can astonish anyone. Indians are associated with top semiconductor companies as CFOs, CTOs, and many other managerial and technological roles.

Despite all the strong pointers, why does India keep missing the bus for setting up a semiconductor fab?

I think what is lacking is an honest effort to bring all the advantages together. We need to work on all the factors and bring in a solution that appeals to possible stakeholders in an economically viable manner. Whether it is land availability, power availability, access to ports, or availability of high quality chemicals and reliable labour, all these need to be coordinated in such a way that players find it a good deal to make semiconductors in India.

> LAM Research is a USA headquartered company that manufacturers equipment needed for semiconductor manufacturing. They have been manufacturing equipment used for making chips since forty years. Their biggest clients include Samsung and Intel among others. Their India office focuses on hardware design, software, and materials procurement.

Today, it is easy to go to Taiwan and get semiconductors manufactured from a design, and a lot of startups are already designing semiconductors here. The investment required to set up a modern fab is so huge that private players might not be ready to invest. So, the government might have to intervene, encourage, and offer as much help as possible.

Why should India have a fab at all?

India has a significant presence in every technology and vertical except electronics, be it medical, software, or some others - you name it and India is among the leaders. However, when it comes to electronics, we have not been able to make our presence count. If you look at China and carefully assess what the country did, you will find that they took calculated moves following a very large electronics import bill and worked on making themselves self-sufficient in terms of electronics.

Twelve years ago, the Indian Semiconductor Association published a report pointing out that the import bill for electronics was very soon going to